A Falsifiable Certificate Architecture for a Physical-Roof Determinant in the Equilateral Three-Disk Flow

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Abstract

This article specifies a falsifiable certificate architecture for a classical transfer determinant driven by the physical Euclidean-flight roof of the no-eclipse equilateral three-disk flow at center separation $d = 6a$. It synthesizes a frozen 26-source corpus and reports no constructed roof, operator theorem, determinant computation, numerical enclosure, fidelity result, nontransfer certificate, or novelty assessment. The evidence supports a strict type firewall among physical dynamics, symbolic suspensions, semiclassical periodic-orbit objects, exact-quantum determinants, and classical transfer determinants. It also supplies conditional components for collision coding, transfer operators, determinant identities, numerical approximation, and Livšic-type obstruction logic, but not the exact object map required for the frozen geometry. We organize the resulting obligations into six open gates: physical-roof construction, operator eligibility, a common coefficient map, a five-channel common-norm error theorem, typed physical-fidelity controls, and theorem-licensed nontransfer. The error contract distinguishes four numerical channels—orbit tail, rank or projection, quadrature or evaluation, and roundoff—from separately propagated geometry and roof-input uncertainty. No channel may be combined without explicit stability, propagation, dependency, and determinant-conditioning arguments. Internal Euler, trace, and Fredholm agreement remains roof-agnostic calibration. All citations lack passage locators, so claim-to-passage support is inconclusive. The fixed state remains `A0_FAIL / A2_NOT_ELIGIBLE / NO_ROUTE_PROMOTION`.

Keywords: three-disk scattering; physical roof; transfer operator; Fredholm determinant; error propagation; certificate architecture

Traditional Chinese Abstract / 繁體中文摘要

本文為無遮蔽、等邊三圓盤流在中心距離 $d = 6a$ 時的物理歐氏飛行屋頂，提出一套可否證的古典轉移行列式證書架構。研究只綜整凍結的二十六項來源，不建構點態屋頂、算子定理、行列式數值、誤差包絡、物理忠實度結果或非轉移證書，也不進行新穎性判定。證據要求嚴格區分物理動力學、符號懸掛、半古典週期軌道、精確量子行列式與古典轉移行列式；相同幾何或編碼不能使不同型別彼此替代。本文將未完成義務整理為六道開放關卡：物理屋頂建構、算子適格性、共同係數映射、五通道共同範數誤差定理、型別化物理控制，以及由定理授權的方向性非轉移推論。誤差契約把軌道尾端、秩或投影、求積或評估及捨入四項數值誤差，與幾何及屋頂輸入不確定性分開傳遞；若缺少穩定性、傳播、相依性與行列式條件控制，便不得宣稱誤差可加、完備或已形成包絡。Euler、週期跡與 Fredholm 行列式的內部一致僅是與屋頂型別無關的校準，不證明物理忠實度。所有引文均缺段落定位，故支持狀態仍不確定；既定路線狀態維持失敗、無資格且不晉級。

關鍵詞： 三圓盤散射；物理屋頂；轉移算子；Fredholm 行列式；誤差傳播；證書架構

1 Introduction, Research Question, and Contribution Boundary

The frozen primary object remains the planar, no-eclipse, equilateral three-disk flow with disk radius `a`, center separation `d=6a`, and Euclidean free-flight time between collisions. The symbolic alphabet is the labeled set $\{1, 2, 3\}$, with consecutive equal symbols forbidden. A prospective oriented owner is a realized primitive cyclic word. Each accepted oriented primitive owner has exactly one primitive-ledger row and ledger multiplicity one: cyclic rotations add no multiplicity; reversal is a separately linked oriented owner and each direction is counted once; disk-label permutations are neither quotiented nor converted into an extra degeneracy factor; and an r-th traversal is recorded only as repetition exponent r of the same owner, never as a new owner. This ledger convention does not fix the determinant repetition weight, which remains a Gate-3 coefficient-theorem obligation. Every accepted code class must carry a witness to one realized physical periodic orbit before it may enter a coefficient ledger. The pointwise roof must be computed from the registered collision geometry, not fitted from inherited periodic totals. None of these owner, multiplicity, realization, or coefficient records has been constructed here.

The research question asks what must be certified before a classical nonconstant-roof transfer determinant can be interpreted as belonging to that exact physical flow. Shared geometry and periodic-orbit vocabulary create a particular risk: physical, semiclassical, exact-quantum, and classical transfer objects can be discussed with similar symbols while carrying different maps, weights, domains, and spectral meanings. The study therefore begins with an object-type firewall rather than with determinant values.

The proposed answer is a six-gate program. A future implementation would have to construct the physical roof, prove operator eligibility, prove a common coefficient map, establish a five-channel common-norm error theorem, validate physical fidelity against typed controls, and license a directional nontransfer inference. Each gate has a stop state, and none is closed in this article. The program is intended to make a later numerical claim falsifiable rather than to imply that a determinant already exists.

The error gate receives special emphasis because a list of numerical tolerances is not an enclosure. Four numerical components are prospectively distinguished: orbit truncation, finite-rank or projection approximation, quadrature or determinant evaluation, and roundoff. A fifth and separately propagated channel is the uncertainty in geometry and the pointwise roof supplied to the operator. These heterogeneous channels may be combined only after a common norm, stability map, propagation rule, and determinant-conditioning bound are proved. The present architecture does not call their sum a total error, claim exhaustiveness, or assume additivity.

The bounded contribution is a prospective integration of a physical-roof input contract, operator and coefficient eligibility gates, a five-channel error theorem template, lawful controls, deterministic replay, and an optional nontransfer module for one frozen classical-dynamics system. The constituent ideas are not claimed as new in isolation, and the frozen corpus is not an exhaustive novelty review. The article reports no roof, operator, determinant, error enclosure, fidelity comparison, cohomology result, or Route credit; internal physical-roof coherence cannot repair absent Route-A arithmetic specificity.

1.1 Typed claim graph and falsifiability

The central discipline is to bind every assertion to a typed object. The physical flow is determined by disk geometry, the collision section, and Euclidean flight time. A symbolic suspension consists of a shift or return map together with a roof; changing that roof changes the timed flow even when the admissible words are unchanged. A semiclassical construction may use the same periodic rays while assigning phase, stability, or wave-number weights. An exact-quantum determinant refers to a different operator and spectral question. A classical transfer determinant again requires its own operator, potential, space, normalization, and analytic domain. Similar notation and a shared orbit vocabulary do not erase these differences.

This graph separates three levels of agreement. Formal agreement means that Euler, repetition, trace, and determinant expressions have been expanded using the same declared weights and conventions. Analytic agreement additionally requires a lawful operator and a theorem relating its traces or spectrum to the determinant on a stated domain. Finite numerical agreement compares declared approximations under an explicit tolerance. Each level can reveal an implementation inconsistency, but none by itself identifies the roof as the Euclidean-flight roof of the frozen geometry. Physical identity requires an independent geometry-to-roof construction and validation against geometry-derived periodic data.

Falsifiability enters through stop states rather than through an expectation of success. If the geometry cannot produce a pointwise roof without fitting inherited totals, Gate 1 stops. If a proposed function space does not satisfy the required operator hypotheses, Gate 2 stops. If orbit and projected-operator coefficients cannot be connected under one normalization, Gate 3 stops. Missing error propagation, physical-control failure, or an unavailable cohomology hypothesis map likewise stops the corresponding gate. These stops prevent an attractive determinant plot or internal identity from compensating for an upstream type error.

The architecture therefore treats provenance as part of the mathematical claim. A determinant value without the roof hash, operator definition, coefficient convention, error contract, and upstream receipts is not an incomplete instance of the registered result; it is a differently specified object. Conversely, an auditable stop does not prove that no suitable roof or operator could ever exist. It records exactly which registered evidence obligation remains unmet. This bounded interpretation is what makes the proposed workflow useful without converting it into a theorem or experiment.

2 Frozen Literature and Theoretical Frame

2.1 A determinant-type firewall

The classical three-hard-disk record anchors physical geometry, trapped trajectories, collision coding, and geometric lengths, but any formula-level use remains bound to correction companion DOI `10.1063/1.457669`; it does not certify a project roof ledger or transfer determinant [12].

The companion semiclassical study organizes periodic-orbit and zeta constructions and is subject to the same correction DOI for affected use, yet its spectral object is not the classical physical-roof determinant [13].

The exact-quantum scattering determinant is a further typed comparator, with affected use bound to correction companion DOI `10.1063/1.457670`; it cannot be substituted for a classical transfer determinant [14].

Three-disk periodic-orbit quantization supports cycle organization in chaotic scattering without proving nuclearity or a project error contract [6].

Periodic-orbit expansions for classical smooth flows provide adjacent organizational machinery but no convergence or coefficient certificate for the frozen roof and cutoffs [7].

A broad treatment of hyperbolic `n`-disk scattering clarifies symbolic and determinant-type distinctions while leaving quantum trace-class conclusions unavailable to an unconstructed classical roof operator [27].

Rigorous work on waves outside several convex obstacles supplies geometric and periodic-ray context but not the project's basis, coefficient window, or numerical tolerance [17].

A primary proceedings record links several-body trajectories, zeta functions, and scattering poles, yet its source type and support boundary do not yield physical-roof fidelity or a certified determinant [18].

These distinctions motivate the registered firewall: common geometry does not transfer credit across differently typed determinants.

2.2 The exact $d = 6a$ object map remains prospective

Symbolic dynamics for hyperbolic flows provides a general interface among a Markov coding, return map, and roof, but it does not identify the exact three-disk collision section or pointwise flight function [4].

General Axiom-A flow and suspension theory gives a broad thermodynamic frame without automatically establishing applicability to the open-billiard repeller under the project's conventions [5].

The closest frozen open-billiard transfer-operator source is conditional on explicit convexity, visibility, normalization, coding, roof, and function-space hypotheses that must be mapped to the exact object [25].

Spectral estimates for Ruelle transfer operators of Axiom-A flows are likewise conditional and do not choose the project's section, regularity class, basis, rank, or cutoffs [26].

Generalized zeta meromorphic continuation provides an adjacent analytic component but no pointwise `d=6a` roof or lawful finite coefficient comparison [21].

Analytic-flow zeta machinery is similarly bounded by its isolated-set and analyticity hypotheses, which have not been discharged for the frozen project object [9].

The smooth-flow zeta record is correction-sensitive: the frozen verification permits its first meromorphicity part but requires the first-party erratum for affected Section-7 use [15].

The erratum is therefore a mandatory boundary for that affected use and does not add physical-roof evidence or independent peer-reviewed support [16].

An exact future object map would need to connect `d=6a` geometry, collision section, symbolic states, inverse branches, pointwise Euclidean-flight roof, roof regularity, potential and normalization, operator space, determinant convention, and complex domain. Shared category labels such as hyperbolic, Axiom A, and open billiard cannot replace field-by-field hypothesis matching. Fitting the roof from inherited orbit totals is excluded because it would reverse the intended physical construction.

2.3 Roof-agnostic internal calibration

Periodic zeta and transfer identities provide the formal background for comparing consistently typed orbit/repetition weights, but physical specificity is not part of that algebraic consistency [22].

Fredholm-determinant machinery can support analytic identities for eligible operators under stated hypotheses, while it cannot establish nuclearity or the physical identity of an unconstructed project operator [23].

The present article treats Euler-product, periodic-trace, and Fredholm-determinant agreement as a design principle with three levels. At the formal level, coefficient expressions can be compared only after their roof, repetitions, signs, and normalization are typed consistently. At the analytic level, a determinant identity additionally requires a specified operator, space, domain, and convergence theorem. At the finite numerical level, agreement is an implementation calibration on a declared window and tolerance. None of these levels proves that the roof is the physical Euclidean-flight roof. Unit, shuffled, neighboring-geometry, and physical roofs may each be internally coherent when constructed consistently.

2.4 Four numerical components and a separate input-uncertainty channel

Explicit transfer-operator eigenvalue bounds can inform a future orbit- or spectrum-tail component only after the operator space and constants are instantiated [1].

Eigenvalue estimates on holomorphic function spaces offer another conditional component, not proof that the three-disk transfer operator acts on the required domain [2].

Spectral-Galerkin analysis for uniformly expanding maps supplies adjacent projection-error ideas but requires a lawful comparison to the flow operator and chosen basis [28].

Numerical Fredholm-determinant analysis supports evaluation-error reasoning for eligible trace-class integral operators while leaving trace class, coefficient correspondence, and project-specific error composition unproved [3].

The prospective error object is therefore a five-entry ledger rather than an asserted scalar identity:

```
E_geometry/roof-input
E_orbit-tail
E_rank/projection
E_quadrature/evaluation
E_roundoff
```

The last four are numerical approximation components. The first is separately typed input or model-enclosure uncertainty. A defensible enclosure would require a declared roof norm, an operator norm, and an output norm; a stability statement from roof perturbation to operator perturbation; a legal coefficient map between orbit and projected-operator representations; propagation of all five bounds through the chosen approximation chain; and a determinant-conditioning estimate on the declared complex domain. Without these links, even individually small terms cannot be called additive, exhaustive, or complete. The frozen corpus supplies no dedicated roundoff theorem, no geometry-input theorem, no common norm, and no determinant-conditioning bound for the project.

2.5 Directional Livšic reasoning

Periodic-data cohomology theory supports a negative obstruction under stated hyperbolic or symbolic hypotheses, but finite periodic agreement cannot prove a global coboundary relation [20].

Regularity and perturbative foundations for the Livšic equation refine the required hypothesis map without replacing all-periodic-data obligations by a finite panel [8].

Matrix-cocycle theory is stronger and differently typed than the scalar roof comparison, so it is background rather than a project-specific nontransfer result [19].

A current abelian extension applies under its own Anosov-flow hypotheses and does not automatically cover the open three-disk repeller or license a positive conclusion from a finite cutoff [24].

The admissible inference is asymmetric. After exact system and theorem hypotheses have been matched, one certified periodic-sum mismatch may obstruct one predeclared relation of the form `rho_1 = c rho_2 + h - h o sigma`, with `c > 0`. Finite agreement does not establish the equality for all required periodic orbits. A mismatch against one scaling relation does not prove every form of inequivalence. No project-specific mismatch or equivalence result is reported here.

3 Executed Methodology: Frozen Evidence Synthesis Only

The executed evidence synthesis remains bounded to the frozen 68 captured manifestations, 52 unique screened records, 26 admitted records, and 24 peer-reviewed-journal records. A dated replay on 3 September 2026 ran all 54 exact frozen query strings through the Crossref REST `query.bibliographic` interface with one top-ranked metadata record returned per query. The row-level ledger records the query, retrieval time, interface, HTTP status, candidate DOI and title, inventory match, screening decision, and reason for every replay row. It retained only candidates that matched the frozen admitted inventory by DOI or normalized title; every other top record was screened out as outside the frozen scope. This replay is a new, fully enumerated retrieval event. The unavailable excluded rows from the original synthesis session remain unavailable and are not reconstructed as historical observations.

Correction provenance is independently citable in the versioned Stage-4-prime bibliography. P30-S01 and P30-S02 are bound to the published correction DOI `https://doi.org/10.1063/1.457669` [10]; P30-S03 is bound to DOI `https://doi.org/10.1063/1.457670` [11]. The existing P30-S17/P30-S18 pairing remains unchanged for its affected Section-7 boundary. These record-level bindings correct publication provenance only; they do not finalize a theorem passage or license any project-specific physical-roof or determinant claim. The canonical bibliography is frozen, and the two entries occur only in the notes-side versioned bibliography used for this revision preview.

Each admitted source was encoded in a source-effect matrix with thematic role, claim fitness, admissible use, prohibited transfer, correction applicability, and locator limitation. Four role-separated assessments—editorial, domain, methodology, and adversarial—were performed in fresh contexts within the same model family before author adjudication. They are therefore distinct fresh-context checks, not independent replications. This reader-facing method classifies and synthesizes evidence only; it creates no scientific artifact, experiment, determinant, enclosure, or result.

The claim-to-passage matrix has 28 rows: one for each of P30-S01–P30-S26 and one record-level row for each correction P30-C01 and P30-C02. All 26 paper-specific source rows retain `anchor:none` and `INCONCLUSIVE` passage support; source identity, DOI resolution, metadata checks, and role coding do not establish theorem-to-claim transfer. The two correction rows are finalized only at publication-record level and bind the affected source IDs; they add no formula or theorem locator. No direct quotation is used. Retraction status remains `NOT_CHECKED`, and source-level conflict-of-interest status remains `UNKNOWN_NOT_AUDITED`.

3.1 From source roles to bounded method claims

The evidence transformation follows the registered claim plan rather than searching for a stronger narrative. Each claim carries a frozen source set and a negative constraint. Physical-scattering sources can motivate geometry and coding only within their correction bindings. Symbolic-flow and operator sources supply conditional components only under their own hypotheses. Determinant and numerical-analysis sources describe what an eligible approximation would require but cannot establish eligibility for an undefined operator. Cohomology sources license only directional reasoning after an exact theorem-to-object map. These roles remain separate rather than being merged into a synthetic theorem.

Metadata verification is not passage verification. All inventory records have verified existence and normalized core metadata within the frozen workflow, yet the manuscript has no theorem, page, section, or paragraph locator for any cited premise. Citation comments therefore record a uniform inconclusive claim-to-passage state. Correction provenance is kept separately: the three Gaspard–Rice records retain their companion-DOI obligations, and P30-S17 remains paired with P30-S18 for affected use. This handling prevents an uncorrected formula transfer but does not amount to a structured retraction or full integrity audit.

The closed-corpus method yields a reviewable architecture and a complete 26-entry bibliography, not a scientific object. It did not generate collision coordinates, evaluate a flight segment, choose a section, define a Banach or Hilbert space, construct an operator, calculate a trace, approximate a determinant, or compare periodic sums. Rephrasing a conditional component in a coherent manuscript cannot advance any gate. This explicit execution boundary allows a later implementation to cite the present work as a design contract while preventing the contract from being mistaken for an implementation receipt.

4 Certificate Method Architecture

4.1 Gate 1: pointwise physical-roof construction

A future Gate-1 certificate must preserve the frozen `d=6a` system and the labeled alphabet $\{1, 2, 3\}$. It must freeze the collision section, inverse branches, geometry evaluator, coordinate and precision conventions, no-eclipse checks, and a pointwise Euclidean-flight roof derived without fitting inherited periodic totals. The owner ledger must identify primitive roots, treat cyclic rotations as one oriented class, retain reversal through `reverse_id`, prohibit an unregistered disk-label quotient, and bind every admitted class to a realized physical periodic-orbit witness and multiplicity rule. The roof's regularity and interval-valued input uncertainty must be serialized. Missing geometry, an incomplete owner decision, or a fitted roof yields `PHYSICAL_ROOF_NOT_EVALUABLE`. No roof, owner ledger, orbit witness, or coefficient has been constructed.

4.2 Gate 2: operator eligibility

The roof must be attached to one explicitly normalized transfer operator acting on one declared function space over one complex domain. Every theorem hypothesis must map to a project field, including expansion or hyperbolicity conditions, smoothness or analyticity, nuclear or trace property, and boundary convention. Missing fields produce `OPERATOR_NOT_EVALUABLE`; shared terminology does not confer eligibility. No operator or function space is selected here.

4.3 Gate 3: common coefficient map

Orbit cutoff and finite-rank projection are differently typed approximations. A prospective proof must identify which complete primitive-and-repetition coefficient through order `N` corresponds to which operator trace, projected trace, and determinant coefficient under the same roof and normalization. Agreement of array indices, truncation integers, or similarly named coefficients is not a legal map. If no comparison theorem is available, the correct state is `CALIBRATION_NOT_EVALUABLE`. No coefficient comparison is proved here.

4.4 Gate 4: five-channel common-norm error theorem

Prospective typed theorem template. Let B be a declared complex Banach space with norm $\|\cdot\|_B$, let $\Omega \Subset \mathbb{C}$ be a compact declared parameter domain, and let $L_{s,\rho} : B \to B$ be the normalized operator associated with a registered roof ρ. The intended output functional is $D_\rho(s) = \det(I - L_{s,\rho})$ on Ω, conditional on a recorded trace-class or nuclearity theorem and coefficient correspondence. A future certificate must name the spaces, norms, maps, hypotheses, dependency graph, and symbolic constants needed to consume the following five raw channels; every presently unknown field is `UNASSIGNED`, not a numerical value:

1. a bound for the physical geometry and pointwise-roof input in a declared roof norm;
2. an orbit-tail bound for the complete orbit-and-repetition representation;
3. a rank or projection bound for the selected operator space and basis;
4. a quadrature or determinant-evaluation bound on the declared complex domain; and
5. a roundoff bound tied to arithmetic precision, algorithms, and stopping rules.

Write the raw bounds as $\varepsilon_{\text{roof}}$, $\varepsilon_{\text{orbit}}$, $\varepsilon_{\text{proj}}$, $\varepsilon_{\text{eval}}$, and $\varepsilon_{\text{round}}$. A valid theorem instance must supply a roof-to-operator constant $C_{\rho\to L}$, coefficient and projection propagation constants, and a determinant condition function $\kappa_D(s)$, all with stated hypotheses and domains, so that a composed statement has the prospective form

$$\sup_{s\in\Omega} |\widehat{D}_\rho(s) - D_\rho(s)| \le \sup_{s\in\Omega} \kappa_D(s)\,\mathcal{C}(C_{\rho\to L}\varepsilon_{\text{roof}}, \varepsilon_{\text{orbit}}, \varepsilon_{\text{proj}}, \varepsilon_{\text{eval}}, \varepsilon_{\text{round}}).$$

The function $\mathcal{C}$ must record shared dependencies instead of assuming independence or summing overlapping channels. If a space, map, theorem hypothesis, constant, dependency treatment, domain, or output tolerance is absent, the sole disposition is `ERROR_CONTRACT_NOT_EVALUABLE`. No constant, tolerance, enclosure, or scientific value is assigned here.

4.5 Prospective composition of the five error channels

The five-entry ledger is a typing device before it is a numerical formula. Geometry and roof-input uncertainty begins with the declared disk parameters, collision coordinates, section conventions, and the interval or exact arithmetic used by the flight evaluator. Its output is a set or bound for pointwise roofs in a stated roof norm. Orbit-tail error concerns omitted primitive cycles and repetitions in a particular periodic representation. Rank or projection error concerns replacement of an infinite-dimensional operator by a finite-dimensional one. Quadrature or evaluation error concerns how the selected finite object is integrated or how its determinant is evaluated. Roundoff concerns the arithmetic operations, precision schedule, and stopping rules actually used. These channels enter at different stages and cannot share a unit until connecting maps are supplied.

A lawful composition would begin with a stability statement that transports a roof perturbation into an operator perturbation on the declared function space. It would then relate the complete periodic representation to the operator coefficients under one roof and normalization. Projection and evaluation maps would carry their own operator or coefficient norms. Finally, a determinant-conditioning statement would convert the accumulated perturbation into a bound for the reported determinant quantity over the declared complex domain. Every constant would be attached to a hypothesis record and a domain of validity. A scalar displayed beside five tolerance labels would not provide these maps.

Dependencies require special treatment. Geometry uncertainty can change every orbit weight coherently, so it may overlap with what a naive orbit-tail estimate treats as coefficient uncertainty. Projection and quadrature errors may share evaluations or basis data. Roundoff may be amplified by the same ill-conditioning that controls determinant sensitivity. Consequently, addition by the triangle inequality would need a common ambient norm and verified coverage, while sharper combination would need an explicitly justified dependency model. The architecture prescribes neither numerical independence nor quadrature of errors. It asks a future proof to state which dependence assumptions, if any, it uses.

The theorem template distinguishes local calibration from a uniform enclosure. Its required fields are `input_space`, `roof_norm`, `operator_space`, `complex_domain`, `operator_map`, `coefficient_map`, `raw_channels`, `dependency_ledger`, `conditioning_functional`, `output_functional`, and `output_tolerance`. Agreement at selected parameters or on a precision ladder may test code paths but cannot fill any theorem field. Conditioning near zeros or spectral features must be included in κ_D. At present the domain, constants, functional tolerance, and all raw bounds are `UNASSIGNED`; hence the template is type-readable but `ERROR_CONTRACT_NOT_EVALUABLE`.

4.6 Gate 5: physical fidelity and typed controls

Gate 5 freezes one primary roof and three deterministic controls on a single callable interface. Scale is fixed by $a = 1$, so the primary geometry is $d = 6$ and $\rho_{\text{phys}} = \tau_6$. The unit control is $\rho_{\text{unit}} = 1$. Let ϕ be the order-three cyclic disk-label map $1 \mapsto 2 \mapsto 3 \mapsto 1$; it preserves the no-repeat adjacency rule and, for the equilateral geometry, defines the symmetry control $\rho_{\text{sym}} = \rho_{\text{phys}} \circ \phi$. The neighboring control uses the exact rational perturbation $\delta = 1/10$, hence $d = 6 + \delta = 61/10$ and $\rho_{\text{near}} = \tau_{61/10}$. Its geometry must independently satisfy the no-eclipse precondition. Comparisons are restricted to

$$\Omega = \{s : 1/2 \leq \operatorname{Re} s \leq 2,\ |\operatorname{Im} s| \leq 50\}, \qquad \eta_c = 1/100.$$

For each control c, a future Gate-4 joint enclosure $[L_c, U_c]$ for $\Delta_c = \sup_{s \in \Omega} |D_{\text{phys}}(s) - D_c(s)|$ yields `SEPARATED` when $L_c > \eta_c$, `NOT_SEPARATED` when $U_c \leq \eta_c$, and `INDETERMINATE` otherwise.

Surface	Preserved	Broken or changed	Admissible interpretation
Primary $d = 6$	geometry, coding, owners, physical timing	none	registered baseline only
Unit $c_0 = 1$	coding and owner policy	pointwise flight-time variation and geometry-derived timing	tests dependence on a non-constant roof, not geometric fidelity
Cyclic ϕ	equilateral geometry, adjacency, owner relation, roof values up to relabeling	no physical property; labels are permuted	symmetry/invariance control for label equivariance
Neighbor $d = 61/10$	disk type, symbolic adjacency when no-eclipse holds, owner policy	exact primary geometry and flight times	local geometry-sensitivity control, not an extrapolation theorem

No control comparison or numerical enclosure has been executed.

4.7 Gate 6: theorem-licensed nontransfer

Gate 6 lies outside the minimum physical-determinant certificate. It becomes active only if, before result access, a record freezes two lawful roofs on the same symbolic system, a positive scale convention, the exact cohomological relation under test, a theorem-to-object hypothesis map, and a periodic-sum mismatch predicate with certified uncertainty handling. Without that record the state is `NOT_ACTIVATED`, which is a disclosed absence rather than a waiver. If activated, a certified mismatch may yield `NONTRANSFER_CERTIFIED` for the predeclared relation; otherwise the terminal state is `NONTRANSFER_NOT_PROVED`. Neither state licenses a universal inequivalence claim, and no Gate-6 execution or result exists.

4.8 Gate dependencies, artifacts, and independent replay

The certificate is a directed acyclic graph with the following complete reader-facing gate map. A hash field always binds both the typed inputs and the emitted output; a missing permission or prerequisite stops the gate and every listed downstream consumer.

Gate	Inputs	Output, receipt, and hash	Uncertainty, consumer, permission/stop
1	frozen geometry and owner policy	pointwise-roof and orbit-owner ledger; G1 receipt; input/output SHA-256	geometry/roof-input; G2–G5; construct only from frozen geometry, otherwise stop
2	G1 output and function-space specification	operator specification and theorem-hypothesis map; G2 receipt; input/output SHA-256	hypothesis/applicability; G3–G5; require exact theorem map, otherwise stop
3	G1–G2 outputs and normalization	common coefficient-correspondence ledger; G3 receipt; input/output SHA-256	coefficient/repetition convention; G4–G5; require one typed coefficient theorem, otherwise stop
4	G1–G3 plus five channel records	domain-specific joint enclosure; G4 receipt; input/output SHA-256	geometry/roof, orbit tail, projection, evaluation, roundoff; G5; require common norm, propagation, dependence, and conditioning, otherwise stop
5	G1–G4 plus frozen controls and thresholds	physical-versus-control comparison; G5 receipt; input/output SHA-256	propagated joint enclosure; registered physical interpretation; require all controls evaluable, otherwise stop
6	separately registered roof pair and theorem map	typed directional nontransfer decision; G6 receipt; input/output SHA-256	theorem applicability and witness; nontransfer consumer only; require separate activation, otherwise `NOT_ACTIVATED`

The closed state vocabulary is `NOT_STARTED`, `PREREQUISITE_BLOCKED`, `EVALUABLE`, `PASSED`, `FAILED`, and `NOT_EVALUABLE`; Gate 6 additionally admits `NOT_ACTIVATED`. Prose cannot promote a state.

Every prospective artifact requires deterministic serialization and a content hash. Replay must reconstruct the primitive-root and cyclic-rotation policy, verify that reversal remains separately linked, confirm that disk-label permutations were not silently quotiented, and bind each accepted code class to one realized `d=6a` periodic orbit before coefficients are consumed. It must then re-evaluate collision geometry, ensure that no periodic total fitted the pointwise roof, verify operator and coefficient metadata, reproduce error arithmetic, and confirm that each control is a lawful pointwise roof. Replay agreement would establish reproducibility of the registered finite procedure, not a primitive-cycle census, determinant result, physical truth beyond the proved scope, or Route credit.

Gates 1–5 are currently `NOT_STARTED`; none is `EVALUABLE`, `PASSED`, or `FAILED`. Gate 6 is `NOT_ACTIVATED`, because no separate roof-pair registration and theorem map exists. After execution begins, an unmet predecessor yields `PREREQUISITE_BLOCKED`; complete typed inputs yield `EVALUABLE`; validation then yields `PASSED`, `FAILED`, or `NOT_EVALUABLE` according to the gate contract. The first nonpassing mandatory gate prohibits every downstream physical-determinant claim. This status report is architectural and promotes no gate.

5 Evidence-Synthesis Findings

The first registered finding is a type boundary. The physical flow, semiclassical periodic-orbit constructions, exact-quantum determinants, and classical transfer determinants remain distinct even when they share geometry or coding. Evidence for one object cannot be transferred to another without an exact map.

The second finding is that the `d=6a` physical-roof certificate requires a complete object map from collision geometry through roof regularity and normalization to an eligible operator space and determinant convention. The corpus supplies conditional components, not that instantiated map. Fitting a pointwise roof from inherited periodic totals remains prohibited.

The third finding is that Euler, periodic-trace, and Fredholm-determinant agreement is roof-agnostic internal calibration for a consistently typed roof. It is not evidence that the roof is physical. Formal, analytic, and finite numerical levels must remain separate.

The fourth finding is the revised error architecture. Four numerical components must be supplemented by separately propagated geometry/roof-input uncertainty. A common norm, stability, propagation, and determinant-conditioning contract is mandatory before any combination. The article does not assert a complete total error or claim that the frozen sources close any of those project-specific obligations.

The fifth finding preserves the directional asymmetry of Livšic-type reasoning: a certified mismatch may obstruct one registered relation after hypothesis matching, while finite agreement cannot establish global equivalence. The sixth finding organizes these boundaries into six open gates. The seventh identifies the article as a falsifiable physical-determinant methods and error-contract contribution, not a computed determinant or arithmetic candidate result. The eighth preserves the Route boundary without change.

6 Reproducibility and Prospective Interface

Reader-auditable Stage-4-prime materials are enumerated in `notes/stage4_prime_reader_artifact_manifest_round2.json` (SHA-256 `9fd3373b636e1f74eded47992dcda9230ae34493b1f289c331f2fe0286570bc5`). Every retained entry carries a schema or format, byte length, full SHA-256 digest, and repository-relative access state. The branch-relative locator is `https://github.com/maris205/hilbert-polya-structure/tree/main/flow_systems/papers/30-three-disk-nonconstant-roof-determinant`. It is not a content-addressed or persistent archival release. The manifest supports only the bounded search replay, screening decisions, correction metadata, and source-role matrix; it lists no scientific result package.

A future scientific package would require deterministic schemas for geometry, roof, operator, coefficients, errors, and controls. They would record `a`, `d=6a`, the section and collision inputs; pointwise evaluator bytes and roof uncertainty; operator space, domain, potential, basis, and determinant convention; orbit-to-trace coefficient bindings; all five error channels with norm, stability, propagation, dependency, and conditioning witnesses; and typed physical, unit, shuffled, and neighboring-geometry roofs.

A fresh verifier would check hashes, types, completeness, precision ladders, and stop states under prospectively frozen rules. Role-separated evidence synthesis and the dated literature replay check provenance and declared boundaries only. No serializer, proof object, scientific implementation, determinant, or replay receipt for Gates 1–6 is claimed to exist.

7 Discussion and Implications

Physical fidelity begins before numerical approximation. Accurate evaluation for an approximate roof can still miss the intended physical determinant when the operator is sensitive to roof perturbations. Geometry uncertainty must therefore travel through explicit stability and conditioning maps rather than being hidden inside quadrature or roundoff.

The frozen controls have distinct interpretations. The unit roof preserves coding and owner policy while removing pointwise temporal variation; it can diagnose dependence on a nonconstant roof but not geometric fidelity. The order-three cyclic label map is a symmetry/invariance control: it preserves equilateral geometry, adjacency, and roof information up to relabeling, so it tests label equivariance rather than placement sensitivity. The neighboring $d = 61/10$ roof changes the exact geometry while preserving the registered symbolic interface only if no-eclipse is separately certified; it can test local geometry sensitivity but cannot alter or extrapolate from the primary $d = 6$ system. All use the fixed Ω, $\eta_c = 1/100$, joint enclosure, and three-way decision rule stated above. No comparison or enclosure has been executed, and the controls create no Route-A or Route-B credit.

8 Acknowledged Limitations

The source corpus is bounded and is not a field-wide novelty review. It cannot support a claim that no exact object map, error theorem, control design, or comparable certificate workflow exists outside the frozen search. The manuscript therefore claims only a project-specific integration and typing of those obligations for the unchanged `d=6a` three-disk system. Source age, corpus composition, and architectural coherence do not establish priority, scientific performance, or comparative significance.

The 28-row matrix retains `INCONCLUSIVE` theorem-passage support for P30-S01–P30-S26 and finalizes only the publication records for P30-C01 and P30-C02. P30-S01/P30-S02 are bound to [10], P30-S03 is bound to [11], and P30-S17/P30-S18 remain paired for affected use. The two new bibliography records therefore close the publication-level correction citation gap, but no formula, theorem, section, paragraph, or quotation is cleared for project-specific transfer. Retraction status remains unchecked, and source-level conflicts remain unaudited.

No pointwise roof, exact section, operator, function space, coefficient map, rank, quadrature, precision ladder, determinant, error bound, physical period validation, control roof, or nontransfer witness is constructed or evaluated. The corpus does not supply a dedicated project roundoff theorem, geometry-input theorem, common norm, roof-to-operator stability estimate, propagation theorem, or determinant-conditioning bound. The five-channel contract is therefore a prospective architecture, not a complete total error.

The source retains nonprinting citation-provenance comments and a compact status ledger because the Stage-2 audit contract requires them. No independently reproducible search supplement was authorized. The article type is fixed as methods and evidence architecture, while contribution originality remains unassessed.

Any later execution package should expose the smallest evidence unit that permits independent failure diagnosis. The geometry layer should carry disk and section conventions, collision points, interval inputs, and pointwise flight evaluations. The operator layer should carry the potential, normalization, space, complex domain, and field-by-field hypothesis map. The coefficient layer should record primitive words, repetitions, trace conventions, projection indices, and the exact comparison theorem being used. The error layer should preserve every raw bound before composition, including its norm, constants, dependencies, and valid domain. The control layer should serialize physical, unit, shuffled, and neighboring-geometry roofs through the same callable interface.

That package should also distinguish validation from selection. A control may reveal that an implementation reacts to roof changes, but it may not be chosen after the physical determinant is inspected. A precision ladder may diagnose roundoff, but it does not replace a roundoff proof or a determinant-conditioning bound. A finite periodic panel may expose a mismatch, but its composition and the relation under test must be frozen before the mismatch is used. Finally, replay should fail if any upstream hash changes, even when the numerical output changes only slightly. These requirements make later progress reportable gate by gate without allowing a downstream numerical success to rewrite the physical object.

9 Future Work

The smallest source step is a separately authorized locator pass for open-billiard operator, determinant, approximation, correction, and scalar-cohomology premises, followed by a roundoff and roof-input propagation strategy. The smallest object step is a field-by-field `d=6a` map from geometry to a pointwise roof and eligible operator.

Only after that map is frozen should a common coefficient theorem and five-channel common-norm proof be attempted. A scholar-confirmed design would then register typed roofs, cutoffs, precision, domains, stop states, and replay rules. Any nontransfer test must remain directional and theorem-licensed.

10 Conclusion

The frozen corpus supports a rigorous program for asking whether a classical determinant belongs to the physical `d=6a` three-disk flow, but it does not supply the determinant. The present article preserves the type firewall, turns roof-agnostic calibration into an explicit design principle, and replaces an under-specified four-part total-error claim with a five-channel proof obligation: four numerical components plus separately propagated geometry/roof-input uncertainty under one norm, stability, propagation, and conditioning contract.

All six scientific gates remain open. The defensible output is a falsifiable methods and evidence architecture, not a roof, theorem, computation, enclosure, fidelity result, nontransfer certificate, or novelty finding. The Route state remains `A0_FAIL / A2_NOT_ELIGIBLE / NO_ROUTE_PROMOTION`, the formal tuple remains `UNASSIGNED`, positive arithmetic A2 remains `0/1`, and Route B remains uninvoked.

The architecture makes a future negative outcome interpretable. A roof-construction stop differs from an operator-applicability stop; an error-contract failure differs from a physical-fidelity failure; and a missing theorem map differs from a certified periodic-sum obstruction. Preserving those distinctions is the present contribution. Until separately authorized implementations and replayable certificates exist, none of the six gates may be treated as partially passed by prose alone.

Accordingly, the minimum physical-determinant certificate consists of Gates 1–5 in `PASSED` state under one hash-linked dependency chain. Gate 6 is a separately activated directional module: `NOT_ACTIVATED` is reported when its pre-result roof-pair, relation, theorem map, and predicate record is absent; an activated module terminates only as `NONTRANSFER_CERTIFIED` or `NONTRANSFER_NOT_PROVED`. At present all gates are `NOT_STARTED`; the manuscript supplies only the prospective decision structure and reports no physical, analytic, numerical, control, cohomological, or Route result.

Author Contributions

Liang Wang is the responsible human author. CRediT-style roles: Conceptualization; Methodology; Writing—review and editing; Project administration. He supplied the conceptual direction, made the author-adjudication decisions, approved the recorded workflow gates, interpreted the bounded evidence synthesis, and accepts final accountability for the manuscript. AI assistance is disclosed separately and is not credited with authorship.

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Ethics Statement

Ethics approval, informed consent, and human-subject review are not applicable. This article concerns mathematical objects, published-source records, and project-owned workflow artifacts; it reports no human participants, intervention, identifiable personal data, or human-related dataset.

Data and Materials Availability

The bounded Stage-4-prime evidence files, their schema or format versions, byte lengths, full SHA-256 digests, and repository-relative access states are listed in `notes/stage4_prime_reader_artifact_manifest_round2.json` (SHA-256 `9fd3373b636e1f74eded47992dcda9230ae34493b1f289c331f2fe0286570bc5`). The listed branch-relative repository path is not a persistent archive. The notes-side versioned bibliography contains independently citable P30-C01 and P30-C02 correction records; the canonical bibliography remains frozen. Original-session excluded rows and theorem-level passage adjudications remain unavailable. No collision data, owner ledger, pointwise roof, operator artifact, determinant value, error bound, control comparison, cohomological witness, or experiment was generated in this revision.

AI-Assistance Disclosure and Verification Limitation

OpenAI Codex, operating in the GPT-5 model family, assisted during the session dated 2 September 2026 UTC; the exact backend snapshot or build was not exposed. Assistance covered organization of the frozen bibliography and source identities, bounded evidence synthesis, drafting, review integration, LaTeX conversion, and deterministic checks of identifiers, citations, references, hashes, and word counts. No AI-authored scientific data, physical roof, operator, proof, determinant, error bound, cohomology result, or numerical experiment is presented. AI assistance is not credited with authorship.

Liang Wang made the recorded stage-gate confirmations and author-adjudicated design decisions. Neither that responsibility nor the present composition implies human full-text or source-passage verification. Verification remains limited to frozen metadata, abstracts or authorized scope notes, source-verification ledgers, the source-effect matrix, and review artifacts. Exact passages were not frozen; all citation anchors therefore remain `none`, and every claim-to-passage assessment remains `INCONCLUSIVE` pending Stage 2.5.

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